Statistical & Mathematical Modeling

Regression, Time Series, ARIMA, and Financial Models

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AGENDA

What we will go over:

- Overview of modeling techniques
- History & evolution
- Model comparisons
- Visualizations for each model
- Pros and cons of the tools
- Industry & Academic relevance
- Recommendations
- Q&A

What is statistical & Mathematical Modeling?

Statistical & mathematical modeling is the process of using mathematical equations and statistical techniques to represent real-world data patterns and relationships.

Key purposes:

- Prediction
- Pattern detection
- Risk assessment
- Decision making

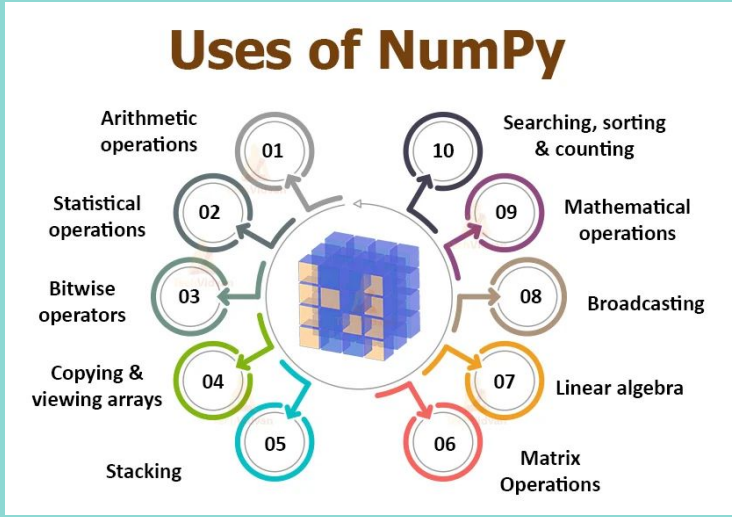
Examples:

- Stock price forecasting
- Sales prediction
- Economic trend analysis

Data Manipulation

NumPy

- Efficiently store numerical data as ndarrays.
- Perform mathematical operations on data.
- Used for T-Test, Chi-Square Test



Array Manipulation	Mathematical	Linear Algebra
np.array() np.reshape()	np.add() np.multiply	np.dot() np.linalg.inv

Data Manipulation

Pandas

- Used for data cleaning, manipulation, and analysis.
- Uses dataframes to work with tabular data.

1

Data Preprocessing

Check for missing values

df.isna().sum

2

Data Cleaning

Remove duplicate values

df.drop_duplicates(subset=[“column_name”])

2

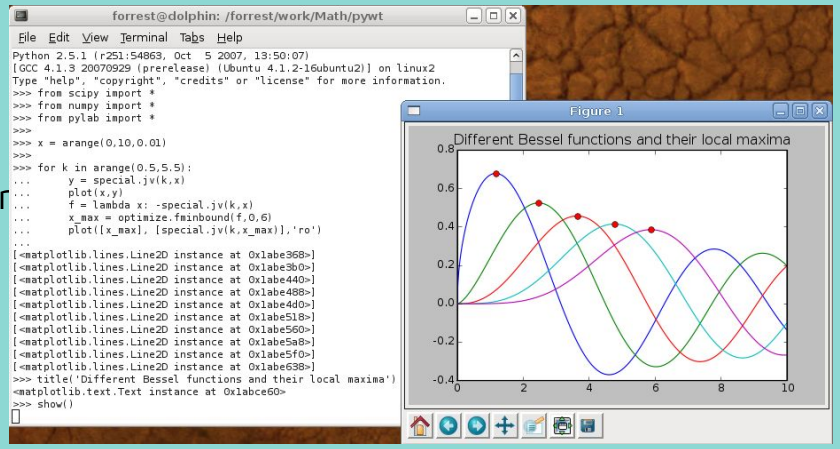
Data Transformation

Filter
Sort
Modify

df.sort_values()

SciPy

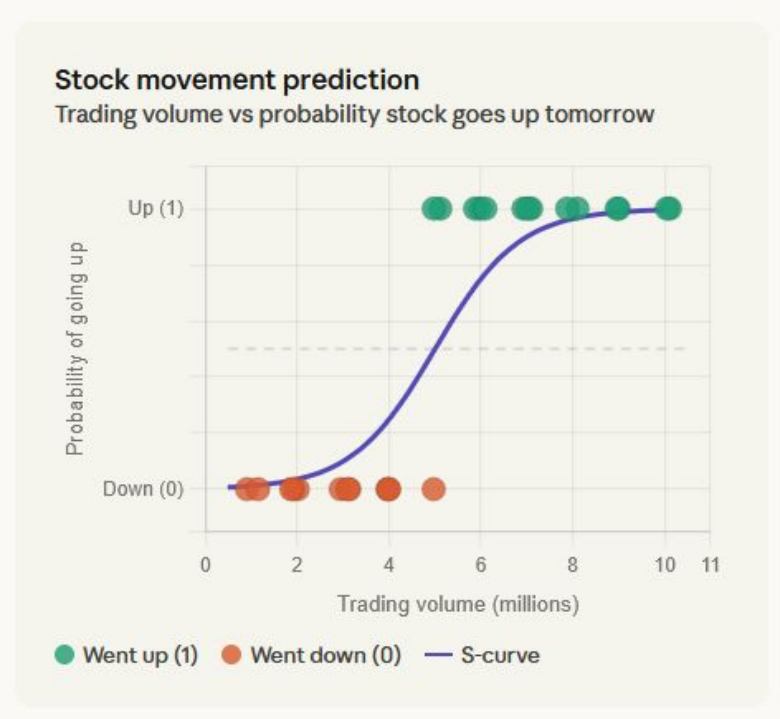
- A scientific computation library built on top of NumPy.
- Solves real world scientific problems and is used for research.
- Supports advanced mathematics and science by providing algorithms and functions.
- Used for T-Test, ANOVA, Mann-Whitney U, Chi-Square tests.



SciPy

Statistical Modeling

	Statsmodels	Scikit-learn
Key purposes	Statistical inference: to understand the relationship between variables.	To build models that analyze data and make accurate predictions.
Best for	Linear & Logistic regression, time series analysis.	Classification, clustering regression, model evaluation, clustering
Pros	Hypothesis testing, provides statistical summaries (p-values, confidence interval, t-statistics)	Machine learning, fast, normal learning curve, nice documentation
Cons	Steep learning curve, not designed for large datasets	Not designed for deep learning (unstructured data); is only effective for tabular data.



Linear Regression - Predict a number based on other variables.

Logistic Regression - Predict probability (for yes or no)

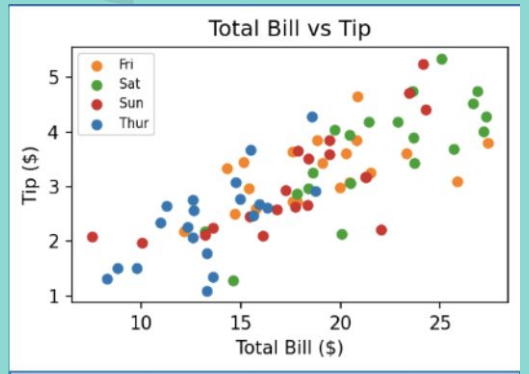
Why Visualization Matters?

- Reveals patterns and outliers in raw data
- Communicates results clearly to non-technical stakeholder
- Speeds up exploratory data analysis
- Essential for model evaluation and reporting
- A core skill for every data scientist and analyst

matplotlib seaborn plotly

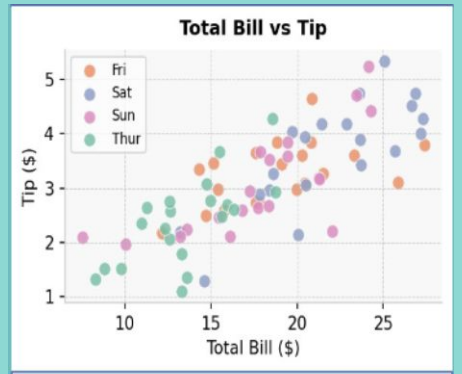
	Matplotlib	Seaborn	Plotly
Key purposes	Custom static charts, publication-quality figures	Statistical EDA, distribution & relationship plots	Interactive web-ready charts, dashboards
Best for	Papers, reports, fine-tuned layouts	Quick EDA, heatmaps, pair plots	Hover/zoom exploration, 3D, maps
Built on	Core library	Matplotlib	D3.js
Pros	Highly customizable, publication-quality	Less code, beautiful defaults	Interactive, 3D & maps support
Cons	Verbose, static only	Less flexible layouts, static only	Heavier, slower on large data

Examples of Visualization



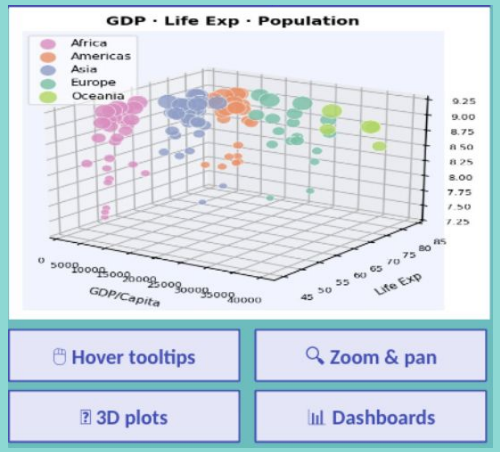
Default styling — works, but plain.
No grid, thick borders, basic colors.

matplotlib



Clean grid, soft colors, bold title,
no top/right borders — instantly polished.

seaborn



plotly

Financial Modeling

What it is

- Uses mathematical and statistical methods to model financial data and relationships
- Analyzes returns, risk, volatility, and investment behavior over time
- Identifies patterns and trends in financial markets to support forecasting and decisions

Purpose

- Forecast future outcomes (prices, growth, performance)
- Measure and manage financial risk
- Compare different investment strategies
- Support data-driven decision making

Where it is used

- Investing: choosing stocks, ETFs, portfolios
- Banking: pricing loans, managing risk
- Portfolio Management: asset allocation decisions
- Business Forecasting: revenue, costs, growth planning

How Financial Modeling Connects

Regression

- Predicts a numeric value from other variables
- Example: estimate returns, prices, or revenue

Time Series / ARIMA

- Predicts values over time using past data
- Example: stock prices, sales, market trends

Financial Modeling

- Applies these models to study valuation, risk, and portfolio decisions
- Focuses on returns, volatility, and forecasting

Visualization

- Scatter plots for regression relationships
- Line charts for time series forecasts
- Risk-return plots for portfolio analysis

Financial Modeling Tools

QuantLib

What is it?
A Python library used for financial pricing and quantitative modeling.

Main use

- Pricing options, bonds, and derivatives
- Modeling interest rates and cash flows

Why it matters

- Helps solve complex valuation problems
- Common in quantitative finance and risk analysis

Q
QuantLib

PyPortfolioOpt

What is it?
A Python library used for portfolio optimization.

Main use

- Allocating money across assets
- Balancing expected return and risk

Why it matters

- Helps build better investment strategies
- Useful in portfolio and ETF analysis

PyPortfolioOpt

Key Outputs in Financial Modeling

Asset Valuation & Pricing

- Estimate the value of financial assets (options, bonds, stocks)
- Based on factors like time, interest rates, and cash flows
- Helps determine if an asset is overvalued or undervalued

Risk Analysis & Uncertainty

- Measure volatility and potential losses
- Evaluate **worst-case scenarios (drawdowns)**
- Understand uncertainty before making decisions

Portfolio Allocation

- Recommend how to distribute money across assets
- Balance **expected return and risk**
- Improve diversification to reduce risk

Forecasting & Market Behavior

- Predict future prices and returns
- Analyze trends using historical data
- Simulate possible future scenarios

Financial Modeling: find the best portfolio given risk and return.

QuantLib → Pandas → PyPortfolioOpt

The chart — all possible portfolios plotted

The output — what PyPortfolioOpt actually tells you to buy

Optimal portfolio weights

Max Sharpe ratio — best return per unit of risk

Apple	32%
Microsoft	28%
Google	22%
Amazon	12%
Tesla	6%

Expected return 17.4%	Risk (volatility) 14.2%
Sharpe ratio 1.23	Portfolios tested 180

Time Series

Data Ordered
Over Time

→

Detect Patterns
Trends & Cycles

→

Forecast
Future values

Pmdarima
Arima-based

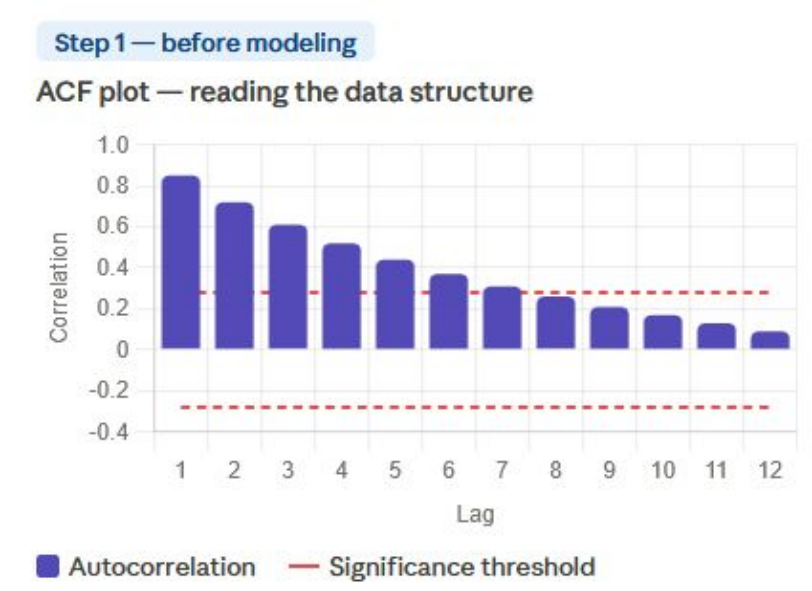
- Automatically finds best model settings
- Best for clean, structured data
- Used in finance & economics
- Mathematically precise

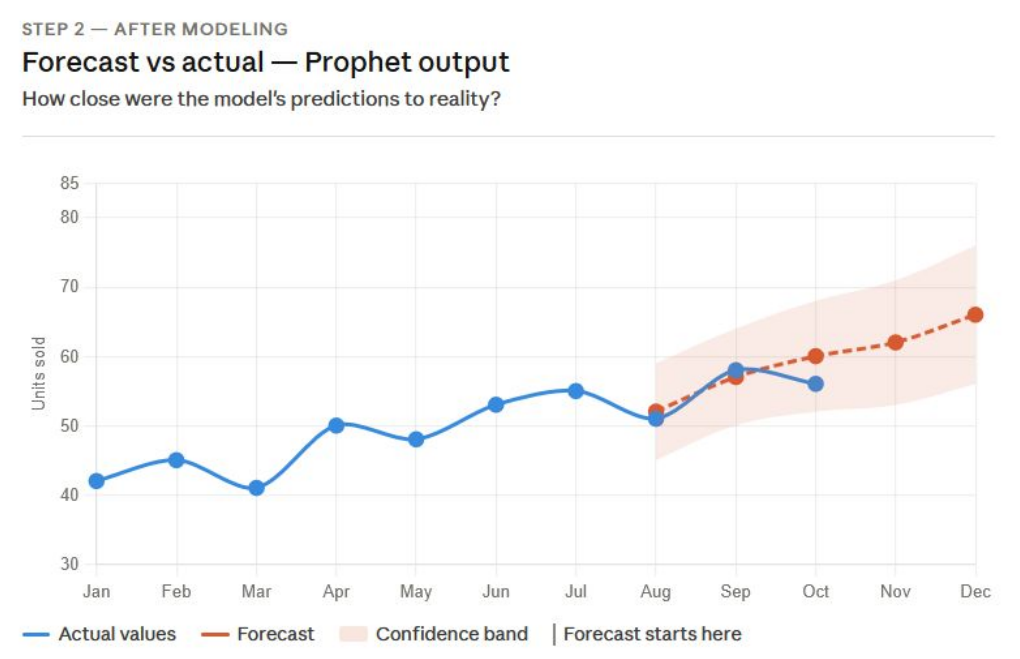
vs

Prophet
Made by Meta/Facebook

- Easy to use
- Best for business/ messy data
- Handles missing data & Holidays
- Used by Meta, retail, startups

Time series: predict a number based on when it is. Learns from the past, forecasts the future.





ACF is manual diagnostic tool with statsmodel/scipy

Real World Use Cases

NETFLIX — *Prophet*

$1B+
Saved Annually

Forecasts Server demand

WALMART — *Pmdarima*

10,000+
Stores Forecasted

Predicts inventory needs

AMAZON — *Prophet*

2.5B
Items forecasted daily

Predicts package volume

INDUSTRY USE CASES

Finance	*Stock & FX forecasting*
Retail	*Demand & Inventory*
Healthcare	*Disease outbreak tracking*
Energy	*power demand prediction*

Questions?

If not, thank you!